

Week 5 — HealthConnect Data Analytics Report

1. Data Preparation

The HealthConnect appointment dataset was prepared for analysis using Python/Pandas during the initial data-quality assessment and Power BI for further transformation and visualization.

The dataset contains **5,000 appointment records and 18 variables**, including appointment characteristics, patient demographics, previous attendance history, reminder information, distance to the clinic, waiting time, and appointment outcomes.

The following data-preparation activities were completed:

- Checked the dataset structure and variable types.
- Converted booking and appointment dates into appropriate date formats.
- Confirmed numerical fields such as age, booking lead days, previous appointments, previous no-shows, distance, and waiting time.
- Checked for missing values.
- Checked for duplicate records.
- Reviewed consistency between related variables.
- Retained legitimate missing values rather than introducing artificial values.
- Used the US date locale where required to correctly interpret the booking dates.
- Applied the cleaned data in Power BI for analysis and dashboard development.

The dataset contained **no duplicate appointment records**. Missing values were mainly observed in distance to clinic, waiting time, and reminder channel. Missing reminder-channel values were expected where no reminder had been sent.

2. Exploratory Data Analysis

The analysis focused on identifying patterns in appointment attendance and no-shows, particularly in relation to patient history, reminders, appointment characteristics, demographics, and operational factors.

Overall appointment outcomes

The dashboard shows that **No-Show is the largest appointment outcome**, representing approximately **48.5%** of all appointments. Attended appointments account for approximately **46.3%**, while cancellations represent approximately **5.3%**.

This indicates that missed appointments are a major operational challenge for HealthConnect and should be an important focus for patient-support interventions.

Previous no-show history

The analysis examined current appointment outcomes against patients' previous no-show history. The dashboard indicates that appointments involving patients with previous missed appointments represent a substantial portion of the dataset.

This suggests that previous attendance behaviour may be useful for identifying appointments that could benefit from additional support.

Reminder status

Appointment outcomes were compared between appointments where reminders were sent and those where reminders were not sent.

The dashboard provides HealthConnect with a way to monitor the relationship between reminder activity and attendance outcomes. However, the analysis is **descriptive** and does not establish that reminders directly caused patients to attend.

Appointment characteristics

Outcome patterns were also examined across:

- Appointment type
- Appointment time
- Appointment day
- Age group
- Gender
- Booking lead time

These comparisons help identify groups and appointment circumstances where missed appointments may be more common.

Distance and waiting time

Average distance to the clinic and average waiting time were compared across appointment outcomes.

These variables provide additional operational context because accessibility and expected waiting experience may influence patients' ability or willingness to attend appointments.

3. KPI Results

The dashboard contains the following core KPIs:

KPI	Result	Business Meaning
Total Appointments	5,000	Overall appointment volume
No-Show Rate	48.5%	Proportion of appointments that were missed
Attendance Rate	46.3%	Proportion of appointments attended
Cancellation Rate	5.3%	Proportion of appointments cancelled
Patients With Previous No-Shows	~2K	Indicates the volume of appointments involving patients with prior missed appointments

The KPIs provide a concise view of HealthConnect's appointment performance and support monitoring of the missed-appointment problem. The Week 5 brief specifically requires 3–5 KPIs to be defined, calculated, and interpreted in relation to business questions.

4. Key Business Insights

Insight 1 — No-shows are a major operational problem

Almost half of all appointments were recorded as no-shows (**48.5%**). This is higher than the attendance rate of **46.3%**, showing that reducing missed appointments should be a significant priority for HealthConnect.

Insight 2 — Previous attendance behaviour is important

A substantial number of appointments involve patients with previous no-show history. This suggests that historical attendance behaviour could be used as an indicator when deciding which appointments may require additional support.

Insight 3 — Reminder activity should be monitored

The dashboard shows different appointment outcomes according to reminder status. HealthConnect can use this comparison to evaluate its reminder process and identify opportunities to improve patient engagement.

Insight 4 — Appointment characteristics influence outcome patterns

Appointment outcomes vary across appointment type, time of day, day of week, and age group. This indicates that missed appointments are not evenly distributed across all appointment circumstances.

Insight 5 — Accessibility and waiting experience deserve attention

Distance to the clinic and waiting time provide useful operational context for understanding appointment outcomes. These factors could be investigated further when designing interventions to reduce missed appointments.

5. Business Recommendations

Based on the analysis, HealthConnect should:

1. **Prioritise patients with previous no-shows** for additional appointment support or follow-up.
2. **Review and optimise reminder strategies**, particularly for appointment groups with higher no-show levels.
3. **Monitor high-risk appointment types and time periods** and investigate why missed appointments are more common in those groups.
4. **Consider accessibility factors**, including distance to the clinic, when designing patient-support strategies.
5. **Monitor waiting times** and investigate whether longer waiting periods are associated with poorer attendance.
6. **Continue using the Power BI dashboard** to monitor no-show, attendance, and cancellation trends and support data-driven decision-making.

6. Limitations

The analysis has several limitations:

- The dataset is synthetic/anonymised and may not represent all real-world HealthConnect patient behaviour.
- Missing values exist for some operational variables, particularly distance and waiting time.
- The analysis identifies associations and patterns but does not establish causation.
- Reminder status cannot by itself demonstrate that reminders caused patients to attend.
- Additional external factors such as transportation availability, patient health status, personal emergencies, and socioeconomic circumstances are not represented in the dataset.

7. Cross-Track Collaboration

For the cross-track collaboration requirement, the **Data Analytics track** can provide the Data Science track with the identified attendance patterns, KPIs, and potential risk indicators.

The key outputs shared include:

- No-show rate
- Attendance rate
- Previous no-show patterns
- Reminder-related outcome patterns
- Appointment characteristic patterns
- Relevant variables that may support further predictive analysis

This collaboration provides the Data Science track with evidence-based analytical findings that can inform future modelling or intervention development. The Week 5 brief requires each track to document at least one meaningful dependency or collaboration point.

8. Week 5 Project Summary

Planned vs Completed

The planned Data Analytics activities for Week 5 included data preparation, exploratory analysis, KPI development, analytical visualization, dashboard creation, business insights, and recommendations.

These activities were completed using the HealthConnect appointment dataset and Power BI dashboard.

Key Outcome

The analysis identified a significant missed-appointment problem, with **48.5% of appointments recorded as no-shows**. The dashboard also provides visibility into attendance patterns across patient and appointment characteristics.

Challenges

The main data-preparation challenge involved correctly interpreting date fields because of date-format differences. This was resolved by applying the appropriate US date locale in Power Query.

Some missing values were also identified in distance, waiting time, and reminder channel fields. These were reviewed and retained appropriately rather than being replaced without justification.

Decisions and Changes

The original Week 4 analytical plan was maintained. The implementation focused on the HealthConnect dataset and translated the planned business questions and KPIs into an interactive Power BI dashboard.